Lab Worksheet 15
IPv4 Addressing - Static Routing

This Lab's Objectives

1. Be able to describe how routers can be configured to route datagrams.
2. Be able to find and fix routing problems.

Expected Outcomes

After the laboratory session, students should be able to:

- configure and view IPv4 static routes in Linux using the *route* or *ip* commands.
- analyse a captured pcap file using wireshark and diagnose simple routing problems across small networks of networks.
- adapt marginally incorrect instructions and / or Netkit configurations.
    - Note carefully - from now on, there may be some deliberate errors introduced so you are not always able to use instructions verbatim from this lab sheet.

Workplan

These lab worksheets assume that you will make accurate notes of everything that you do for later consideration and revision. You should use your lab diary to note down items such as commands used, the outcome of each command, and answers to questions posed in these instructions.

All lab machines have CherryTree Notes software installed, which facilitates organised note-taking. Alternatively, you could buy a hardback A4 laboratory notebook.

NB this is a complex lab – please read each instruction carefully – if you miss out any steps, then don't expect to achieve the correct results! You will need to have completed and thoroughly understood all the steps before the next lab session.

Remember that there is a lab work wiki available to you on BB. Please use this to share and communicate with your colleagues.

Make sure you have completed all the steps in this worksheet before the next lab session.

Lab Worksheet 15
IPv4 Addressing - Static Routing

Starting assumptions

1. You have completed all the work in previous lab sheets.

   - You have used Netkit *lstart* to launch a coordinated group of several virtual machines. Specifically, you are aware of how this group is coordinated via the *lab.conf* file.
   - You know how to shut down virtual machines using *lhalt* and *lcrash*.
   - You have used *ip* and *ping* within the virtual machines.
   - You have saved captured network traffic from a virtual machine using *tcpdump*.
   - You have viewed a packet capture file in the real Linux host using *wireshark*.

Activities

Reminder

- You need to make accurate notes of what you do.
- You will complete unfinished activities before the next timetabled lab session.
- You will resolve what you do not understand by conducting your own careful (and ethically sound) experimentation and / or further reading.

Getting Started

2. Create a directory for lab 15's work. Use a command similar to

```
mkdir -p ~/ctec1704/lab_15
```

3. Download the zipped archive *netkit_lab_15.tar.xz* from Blackboard.
   - This contains the configuration information for 15 virtual machines.
   - The machines are divided into two files *netkit_lab_15_a* & *netkit_lab_15_b*.
   - *Lab a* contains the W and X subnets and part of the D subnet.
   - *Lab b* contains the Y and Z subnets and the remainder of the D subnet.

4. Extract the contents of the archive file, and make sure that all the files and directories for the lab are in the new *lab_15* directory.

```
.../lab_15$ tar -xvf netkit_lab_15.tar.xz
```

5. Place yourself into the *netkit_lab_15_a* directory (there may be a subdirectory netkit_lab_15) and list the contents.

```
.../lab_15$ cd ~/ctec1704/lab_15/netkit_lab_15_a
.../lab_15$ ls -l
```

6. You will see seven directories (some empty, some with content), one for each virtual machine.

Linux Network Interface file

Netkit has the ability to insert files into a netkit vm terminal as it is created. This is the purpose of the machine-named directories in the lab directory, eg any content such as sub-directories or files in the *gwW* directory will be copied into the equivalent directory in the *gwW* netkit vm terminal.

Different distributions (aka distros/versions/flavours) of Linux store this information in different locations in the file system. You need to establish the exact version to know where this information is stored

7. To display information about a Linux system (run this command in a Netkit vm later):

```
.../netkit_lab_15_a$ lsb_release -a 2>/dev/null
```

The Netkit VMs run a recent version of Debian (similar to, but not the same as Ubuntu). In Debian, the *etc/network/interfaces* file contains the static configuration settings for the NIC ethernet interfaces for a Linux computer.

8. Inspect the content of the *gwW/etc/network/interfaces* file:

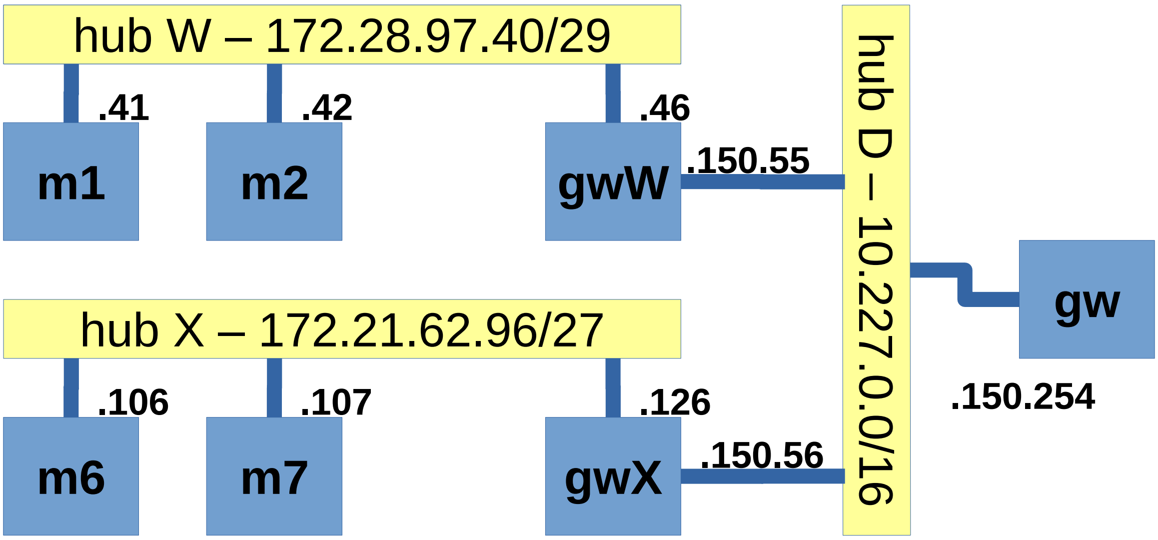
```
.../netkit_lab_15_a$ cat gwW/etc/network/interfaces
```

9. There are three interfaces shown: *lo*, *eth0* and *eth1*. Make a note of the IP addresses for the two *eth?* interfaces.

10. When you get a few minutes later, read more about Linux interfaces in a netkit VM: NB this command only works in Debian.

```
root@m1:~# man interfaces
```

11. Inspect the contents of the *lab.conf* file. Compare the IP addresses for gwW in the diagram with settings in the interfaces file. Confirm that the settings correspond to the diagram below:



12. Look at the seven *.startup files (one for each machine).
   - Identify which of these directly set the IP address via *ip addr* and which use *ifup* to utilise the settings in */etc/network/interfaces.*

13. Each of the gateway machines has several routes pre-configured, eg, for *gwX*, the routes are (in *gwX.startup*):

   ip route add 172.28.97.40/29 via 10.227.150.55
   # ip route add 172.21.62.96/27 via 10.227.150.56
   ip route add 172.19.78.0/23 via 10.227.150.58
   ip route add 172.17.32.0/20 via 10.227.150.58

   ip route add default via 10.227.150.254

   Explain what effect each command will have, especially the final command.

14. Launch the virtual machines using the instruction:

```
.../netkit_lab_15_a$ lstart
```

Look at the configuration

15. Once all machines are up, use *tcpdump* on machines *m2*, *m7* and *gw* to store captured traffic in the files *w4-m2-dump1.pcap*, *w4-m7-dump1.pcap* and *w4-gw-dump1.pcap* in your *captures* directory. Each instruction will be similar to the one below.

```
tcpdump -s0 -i eth0 -w /hosthome/ctec1704/captures/w4-m2-
dump1.pcap
```

16. Record the exact instruction you used on each machine.

17. Now generate some network traffic.
    - From machine **m1** , ping each of the following IP addresses precisely twice:

```
root@m1:~# ping -c 2 172.28.97.46
root@m1:~# ping -c 2 10.227.150.55
root@m1:~# ping -c 2 10.227.150.56
root@m1:~# ping -c 2 172.21.62.126
root@m1:~# ping -c 2 172.16.62.106
root@m1:~# ping -c 2 172.21.62.99
```

18. For each of these IP addresses, write down:
    - which machine it is associated with
    - the corresponding MAC address associated with each

19. Stop the packet capture in *m2*, *m7* and *gw*
    - ie use CTRL-C in each virtual machine in turn.

20. Look at each of the three packet capture files in turn. Explain the MAC address and IP address of each packet.
    - In particular, identify those packets where the destination MAC address is not consistent with the destination IP address that you noted above.

Tracing the Route

21. It is often useful to find the journey that a packet takes as it traverses networks of networks. Start packet capture again on *m2*, *m7* and *gw* but this time capture it into three different *...dump2.pcap* files.

22. From machine *m1*, execute the following commands:

```
root@m1:~# ping -c 1 172.21.62.106 -R
root@m1:~# traceroute -n 172.21.62.106
```

23. Stop the three packet capture files and look at them in wireshark. Explain:
    - how *ping ... -R* works
    - how *traceroute* works
        - why use the -n option?

Fix the Errors

24. Write down the exact instruction you used on each machine.

25. Leave the first set of machines running which you started from the *netkit_lab_15_a* directory ie hubs **W** and **X** and some of hub D.

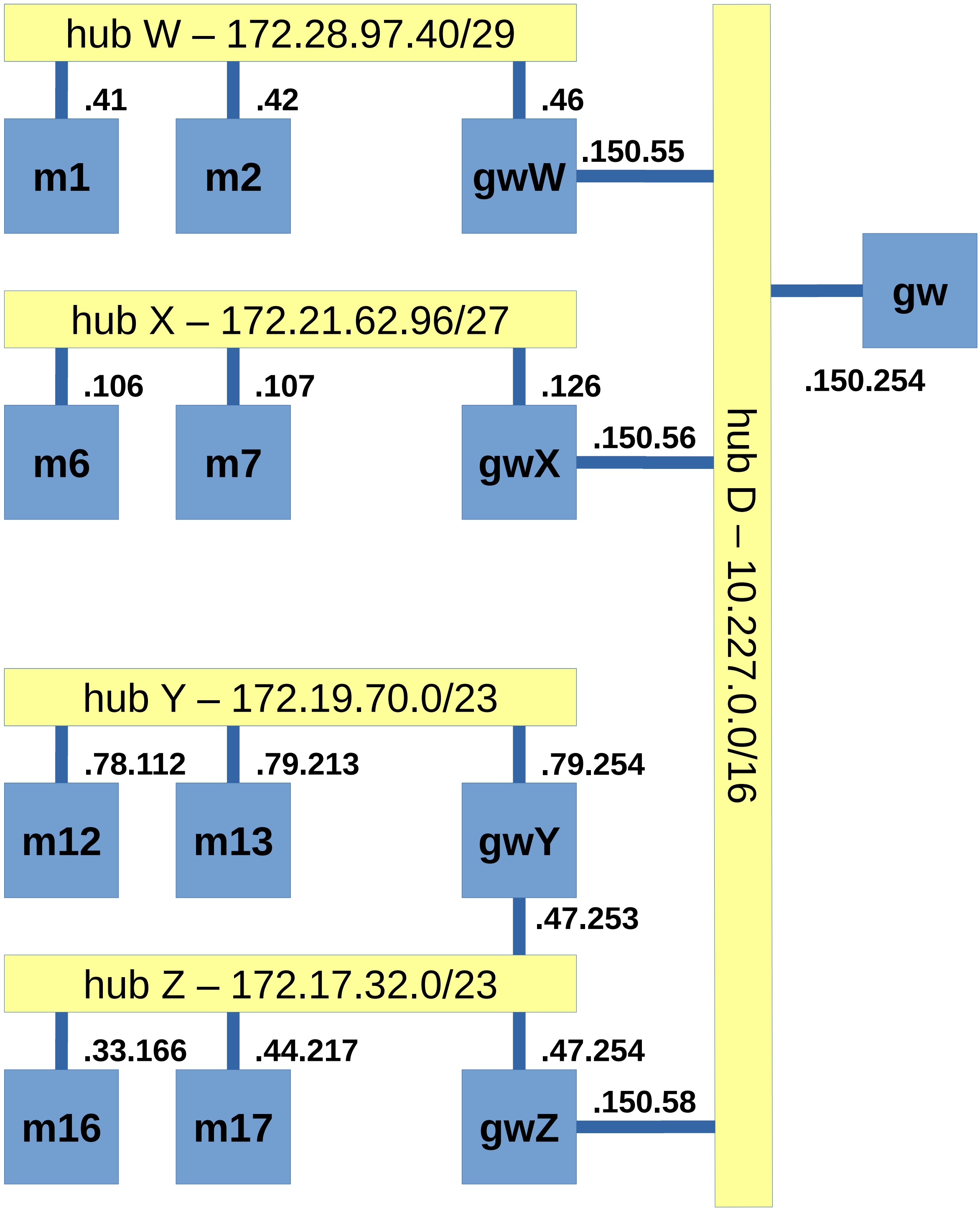
    You are going to open the second lab, to extend the first.

26. Remember that Debian & Ubuntu provide multiple desktops, aka workspaces.

27. Select a new workspace (quickly switch between them using *ctrl alt up-arrow* or *ctrl alt down-arrow*, but doesn't seem to work in the VM) and in it open a terminal with the current directory set to *ctec1704/lab_15/netkit_lab_15_b*

28. Start the netkit lab in the *netkit_lab_15_b* directory using *lstart*.

    - this lab will also use hub D to extend the network to become the overall configuration shown below, plus a Domain Name Server (*dns1*) on hub D which is not shown on the diagram.


hub W – 172.28.97.40/29
.41
.42
.46
m1
m2
gwW
.150.55
hub X – 172.21.62.96/27
.106
.107
.126
m6
m7
gwX
.150.56
gw
.150.254
hub D – 10.227.0.0/16
hub Y – 172.19.70.0/23
.78.112
.79.213
.79.254
m12
m13
gwY
.47.253
hub Z – 172.17.32.0/23
.33.166
.44.217
.47.254
m16
m17
gwZ
.150.58

29. First, find and fix all the IP address and routing problems in this lab so that each machine can ping each other machine. Draw a grid (in your notebook) and tick off each machine pair that can ping. For each fault:

- record the symptoms that you see
- explain what you did to locate what was wrong
- identify what was wrong & note what was wrong
- identify what you did to correct the fault
- record what you did to confirm the symptoms had gone

30. Once the faults are fixed, on machines *m2*, *dns1*, *m17* and *m13* start a packet capture into the *captures* directory in your home directory. Use suitable file names in keeping with the previous pcap files.

31. On machine *m1*, execute:

```
root@m1:~# traceroute -n 172.19.78.112
```

32. Stop the packet captures and use wireshark to explain what is happening.

RFC 1918

33. Look at rfc1918. Wikipedia has a reasonable summary here.

34. Write down the lowest and the highest addresses in each of the private network ranges. Consider what this means for "globally unique IPv4 addresses".

Extension Activities

35. Try creating an additional router between *W* and *Y* (eg give *m13* an extra network card and connect it to hub *W*). Now try to create a set of routes so that a packet to an unknown host will circulate until its time to live (TTL) drops to zero.

36. Make one or two changes to the lab configuration, then challenge a colleague to find and fix them.

37. Experiment with *wireshark's* visualisation of packets.
    - In particular, try *filtering* so you see only the packets you want to
      eg only those packets with a source IP address of 172.28.97.46

Make sure you have completed all the steps in this worksheet before the next lab session.